Detecting Attacks Using Splunk and Sysmon in a Virtual Lab

1. Project Overview

This project demonstrate how security monitoring tools can detect malicious activity within a controlled lab environment. A simulated attack was launched from a Kali Linux machine against a Windows system. Logs generated by Sysmon were ingested into Splunk to monitor and analyse attacker behaviour.

The goal was to understand how Security Operations Centers (SOCs) detect threats through log analysis and event monitoring.

2. Lab Environment

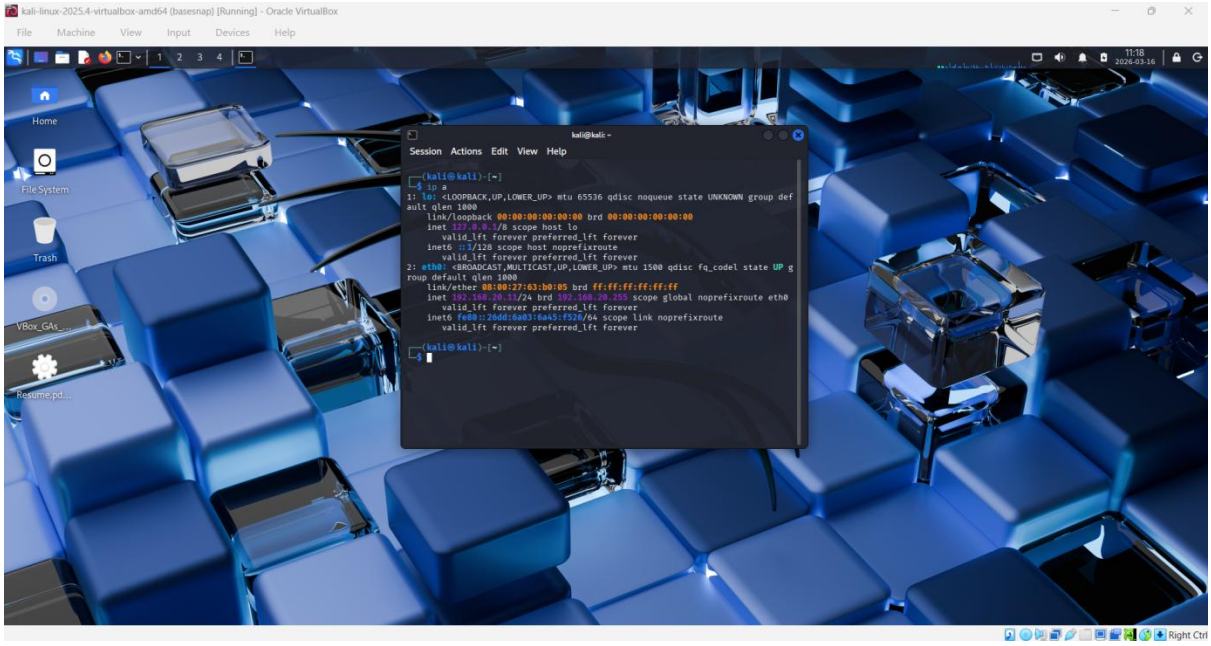
The lab was built using **Oracle VirtualBox** with two virtual machines connected through an **Internal Network** to simulate an isolated enterprise environment.

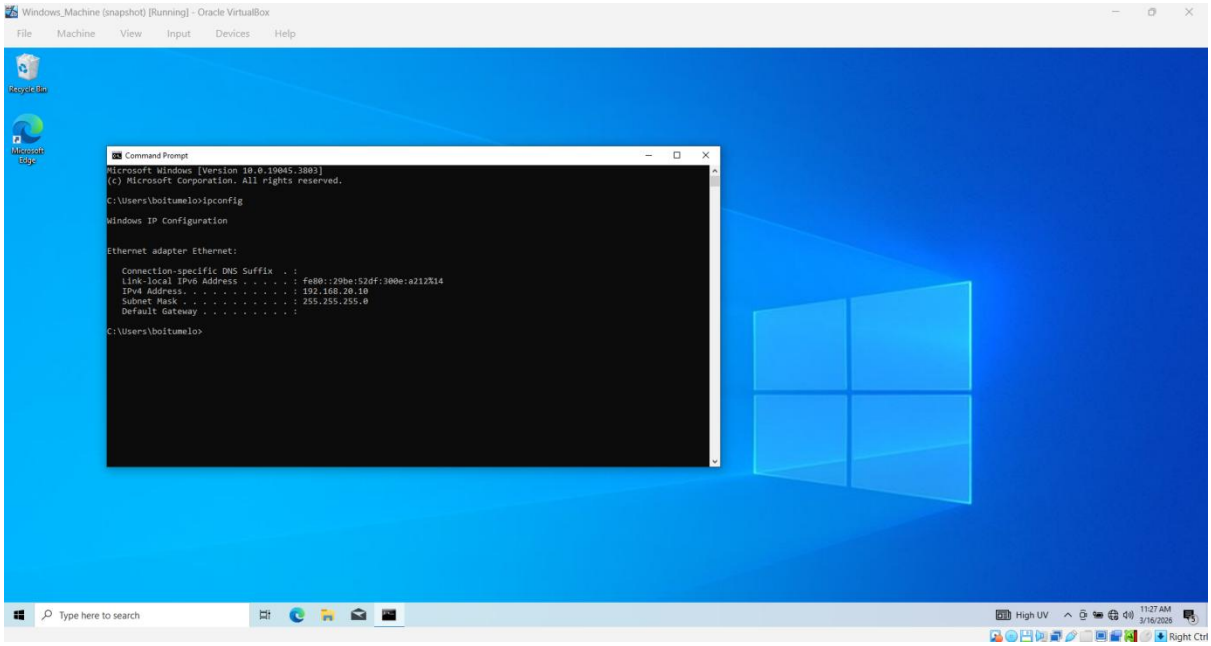
Virtual machine

Kali Linux – Attacker machine

Windows VM – Vulnerable machine

The internal network allowed communication between the machines while keeping the lab isolated from the host network.





3. Tools and Technologies Used

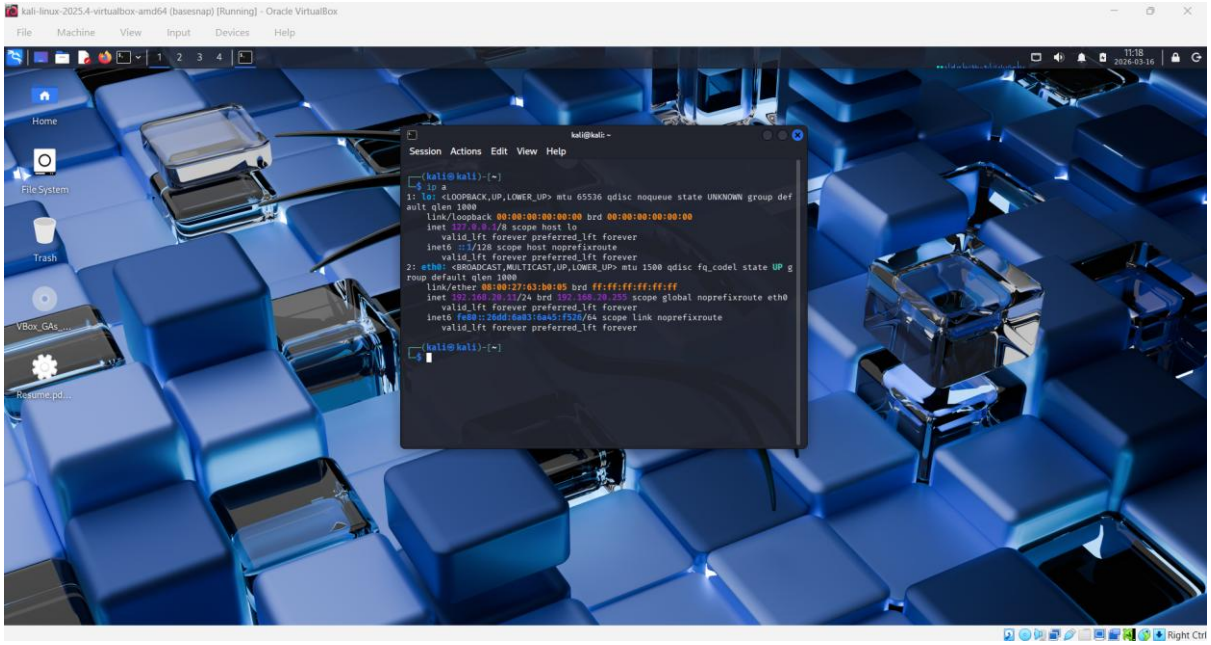
- VirtualBox
- Windows Virtual Machine
- Kali Linux
- Splunk Enterprise (SIEM)
- Sysmon (System Monitor)
- Metasploit Framework
- Python HTTP server

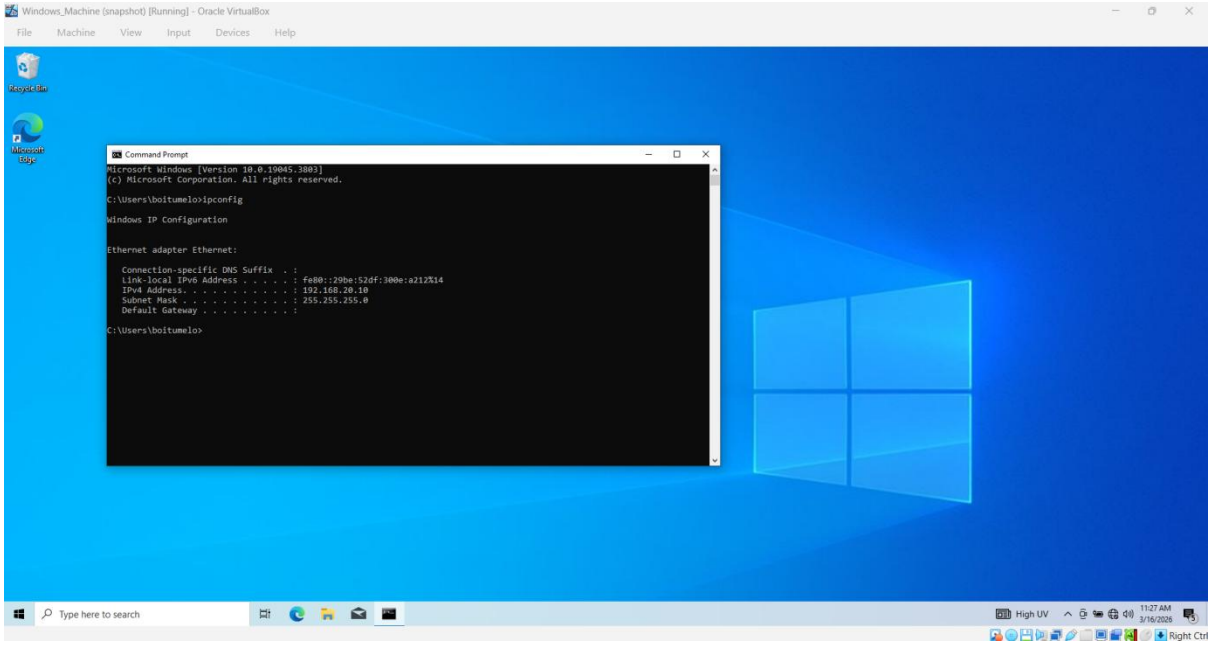
4. Network Configuration

Both virtual machines were configured to use the VirtualBox Internal Network.

This configuration allowed:

- Communication between Kali and Windows
- Safe attack simulation
- Isolation from physical network
- Kali linux IP: 192.168.20.11 and Windows IP: 192.168.20.10



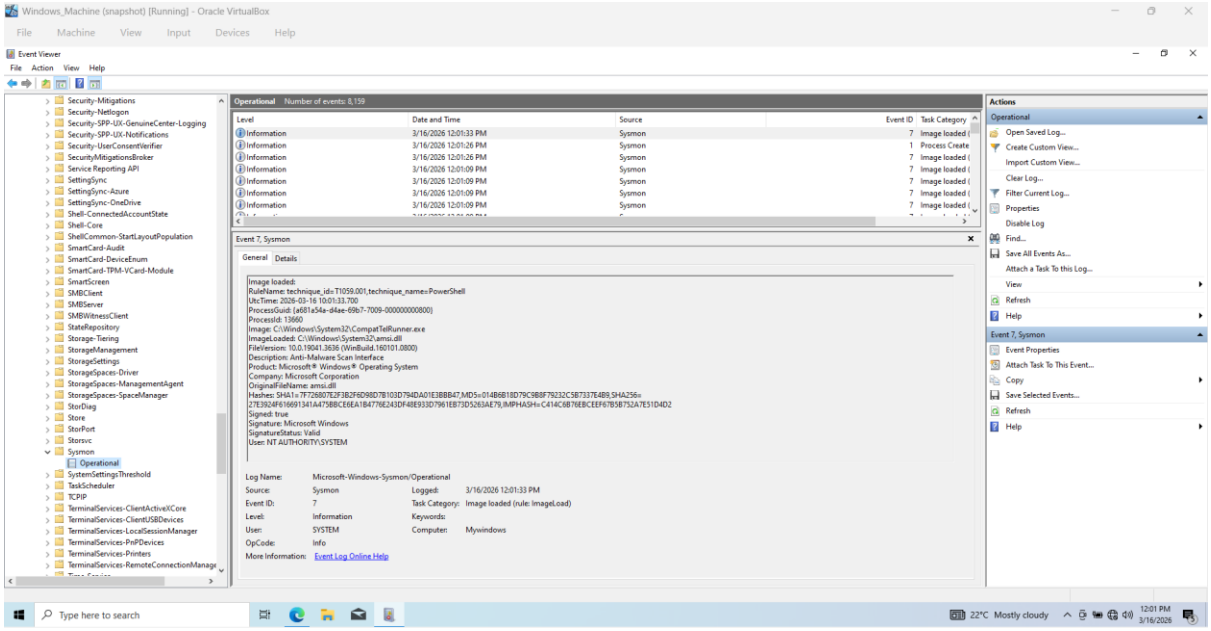


5. Installing Monitoring Tools on Windows

Sysmon Installation

Sysmon was installed on the windows machine to capture detailed system activity such as:

- Process creation
- Network connections
- File changes
- Registry modifications

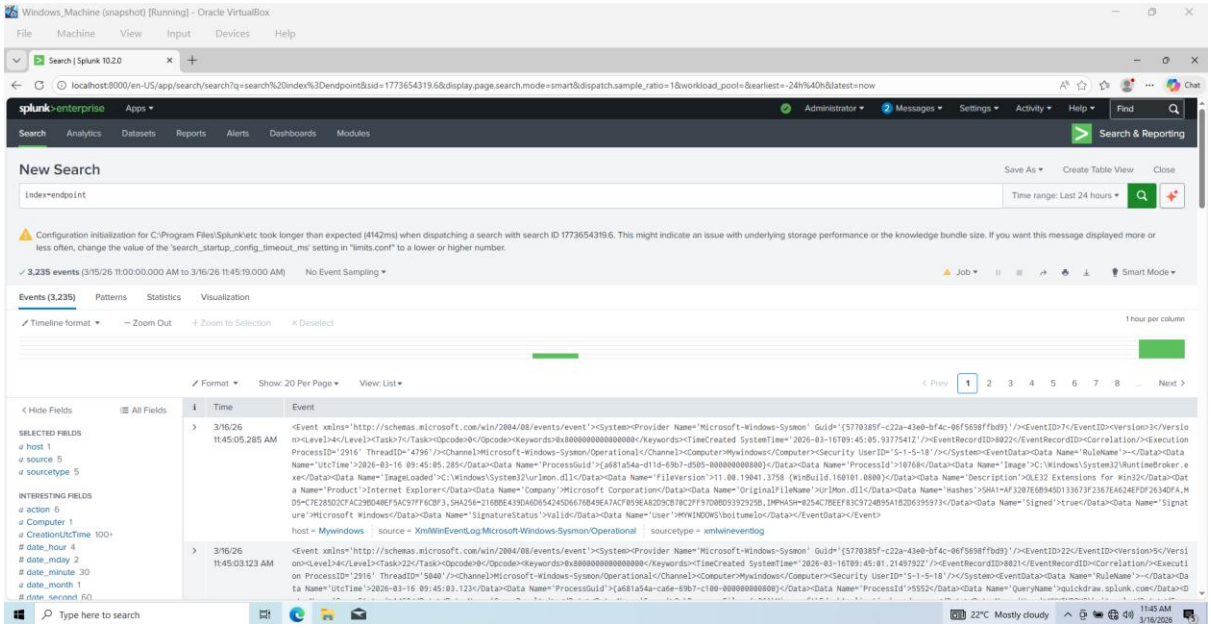


Splunk Installation

Splunk Enterprise was installed on the windows VM to collect and analyse logs generated by Sysmon.

Splunk Web Interface: http://localhost:8000

Sysmon Logs were ingested into Splunk to enable security monitoring and threat detection.



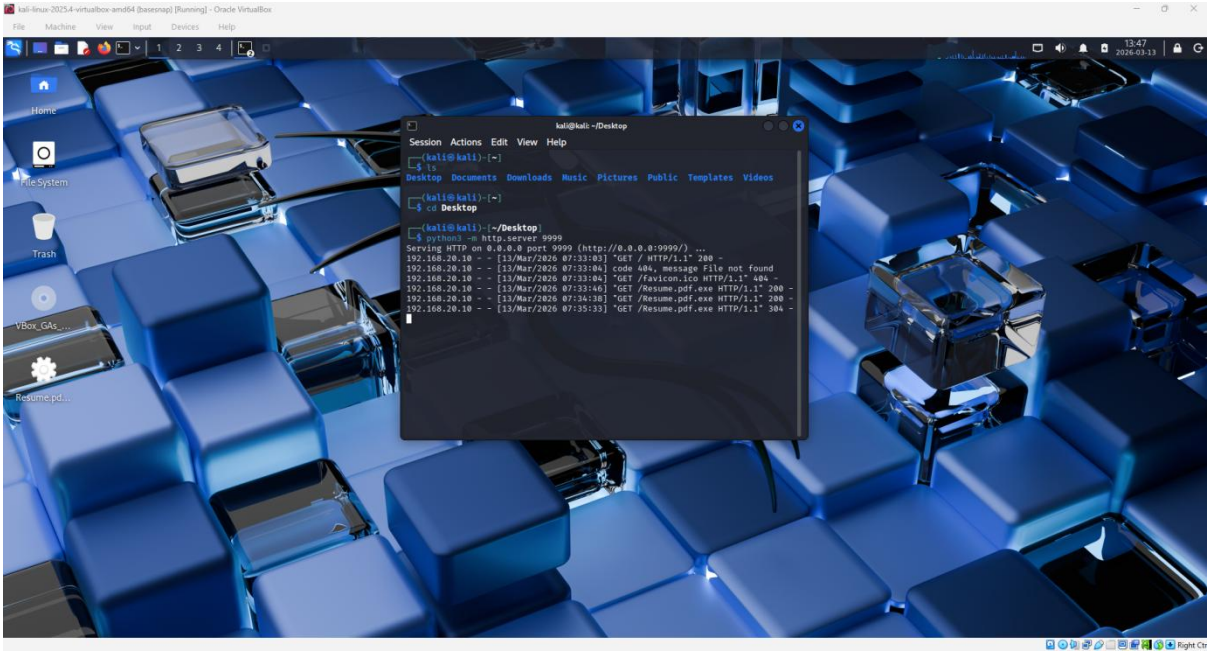
6. Simulating the Attack

To simulate real-world cyber threats, an attack was launched from Kali Linux against Windows machine.

Python HTTP Server

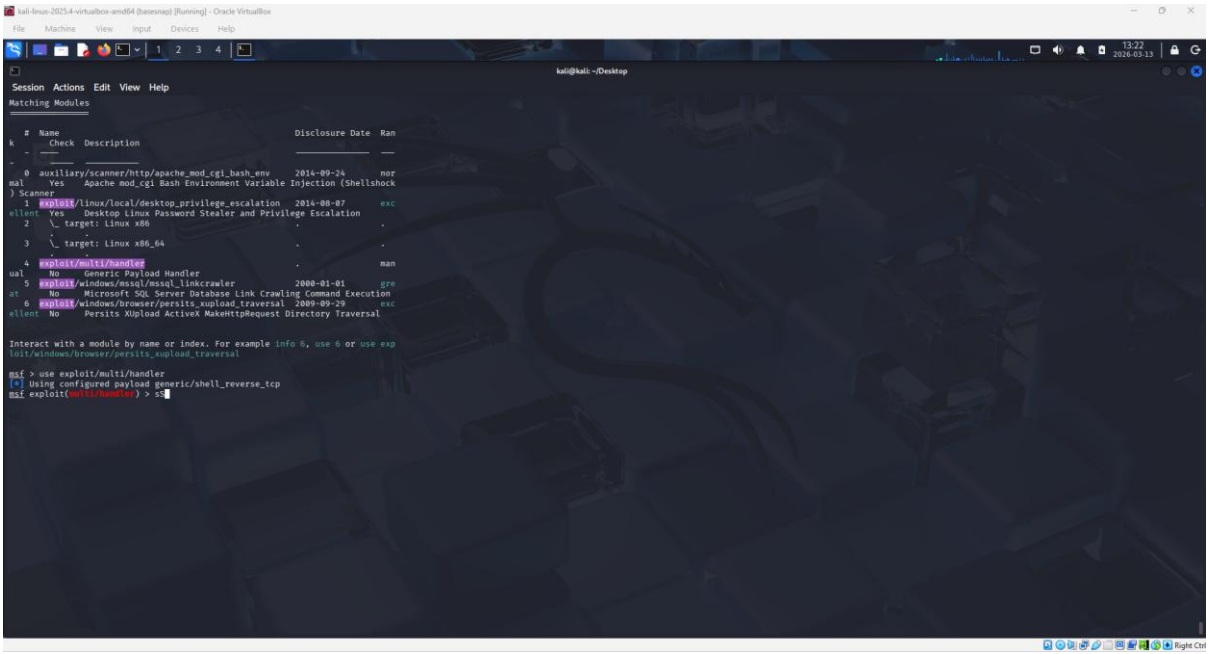
A Python web server was also launched on Kali Linux:

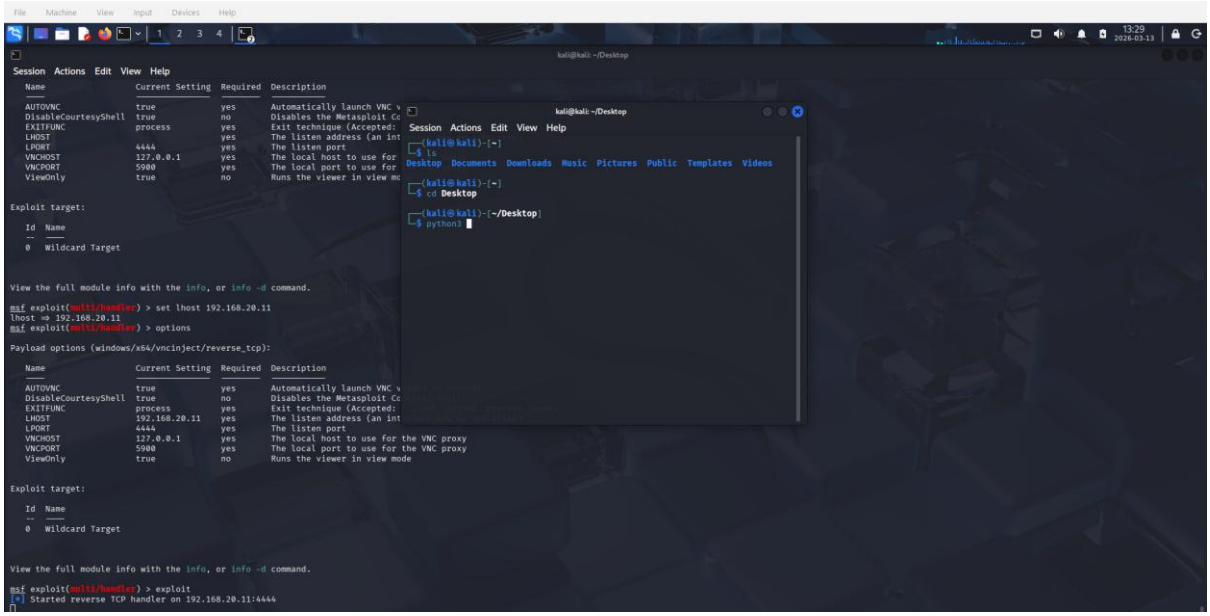
Python3 –m http.server 9999



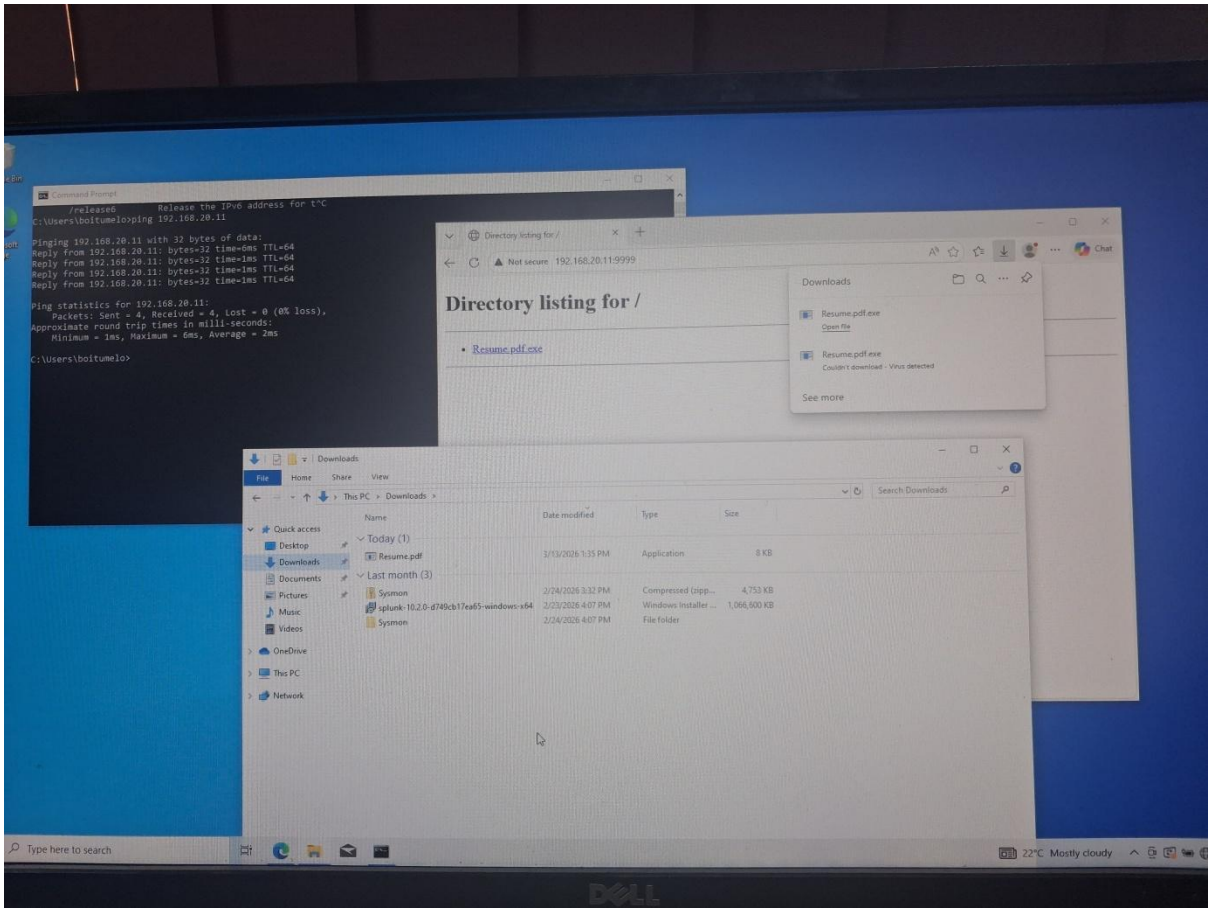
Metasploit Attack

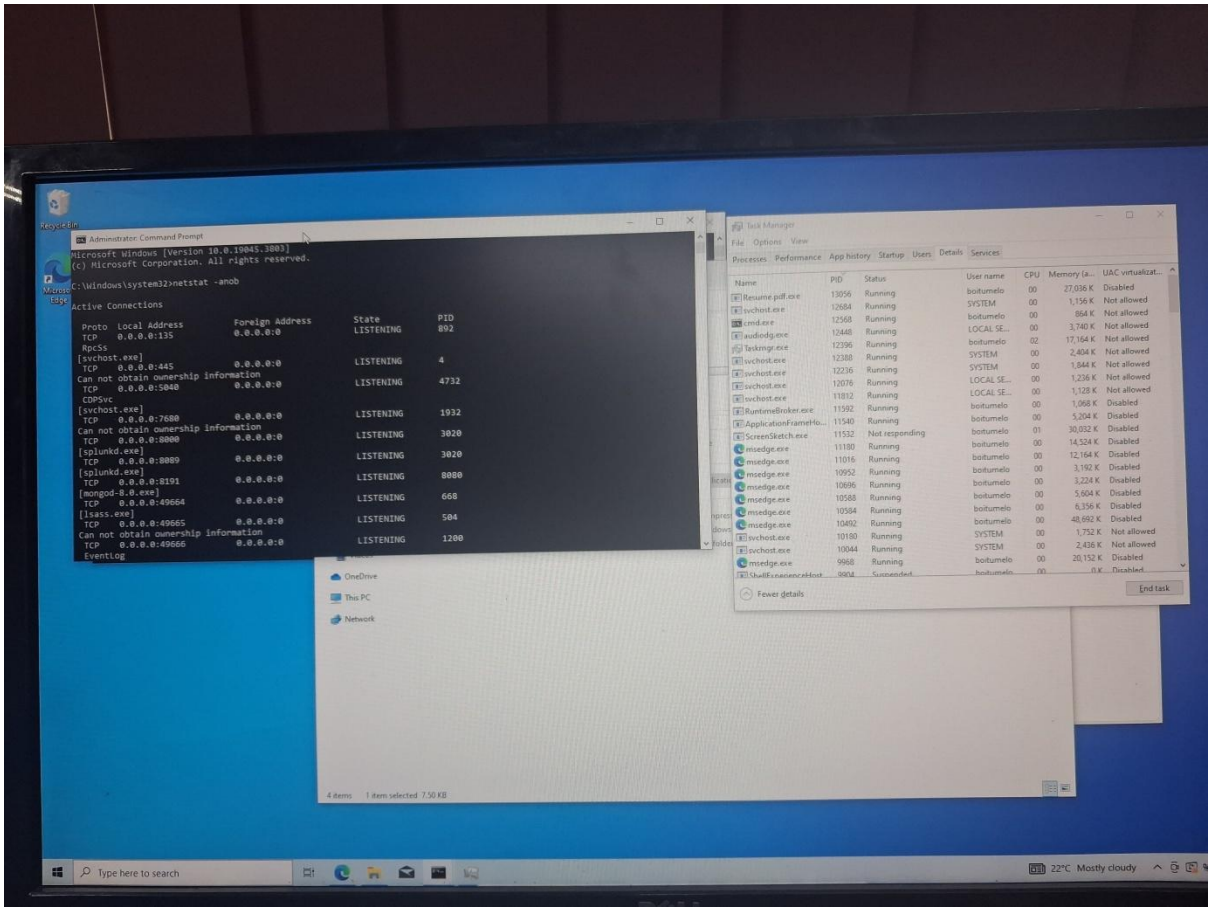
Metasploit was used to generate and execute a payload targeting the windows machine.





The attack created malicious activity on the windows system that could later be created in the logs.





7. Key Findings

During the attack simulation, the following activities were detected:

- Malicious processes created by the Metasploit payload
- Network connections between Windows and Kali attacker machine
- File transfer activity from the Python HTTP server

Sysmon successfullu recorded these events and Splunk enabled visibility into the attack behaviour.

8. Skills Demonstrated

This projects demonstrates cybersecurity skills:

- SIEM log analysis
- Threat detection using Sysmon logs
- Attack simulation with Metasploit
- Security monitoring with Splunk

- Network lab configuration using VirtualBox

9. Conclusion

This project highlights how security monitoring tools such as Sysmon and Splunk can be used to detect malicious activity in a network environment. By simulating real world attacks, security analyst can better understand attacker techniques and improve detection capabilities.

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